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Laporan Akhir

Praktikum Jaringan Komputer

Konfigurasi Vlan, OSPF Multivendor

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1 Langkah-Langkah Percobaan

1.1 Konfigurasi Cisco Switch

Percobaan pertama dilakukan pada Cisco Switch untuk menyiapkan segmentasi jaringan berbasis VLAN. VLAN digunakan agar beberapa jaringan logis dapat berjalan pada satu perangkat fisik, sedangkan trunk digunakan agar banyak VLAN dapat melewati satu link menuju router.

1. Masuk ke terminal Cisco Switch melalui mode privileged dan mode konfigurasi global.
2. Membuat VLAN 10, 20, 30, dan 40 dengan nama Finance, HR, IT, dan Marketing.
3. Melakukan verifikasi menggunakan perintah `show vlan brief` untuk memastikan seluruh VLAN sudah aktif.
4. Mengatur port menuju PC sebagai access port dan port menuju router sebagai trunk port.

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gi1/2, Gi1/3
10	Finance	active	Gi0/2, Gi1/0
20	HR	active	Gi0/1
30	IT	active	Gi1/1
40	Marketing	active	Gi0/3
99	VLAN0099	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
Switch#
```

Gambar 1: Topologi praktikum VLAN, trunk, dan OSPF multivendor

```
Switch#conf t

interface gi0/2
  description PC-FINANCE-VLAN10
  switchport mode access
  switchport access vlan 10
  no shutdown
exit

interface gi0/1
  description PC-HR-VLAN20
  switchport mode access
  switchport access vlan 20
  no shutdown
exit

interface gi1/1
  description PC-IT-VLAN30
  switchport mode access
  switchport access vlan 30
  no shutdown
exit

interface gi0/3
  description PC-MARKETING-VLAN40
  switchport mode access
  switchport access vlan 40
  no shutdown
exit
```

Gambar 2: Pembuatan VLAN pada Cisco Switch

```
Switch#conf t

interface gi0/0
  switchport trunk encapsulation dot1q
  description TRUNK-TO-CISCO-ROUTER
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30,40
  no shutdown
exit
```

Gambar 3: Verifikasi VLAN pada Cisco Switch

1.2 Konfigurasi Cisco Router

Pada percobaan kedua, Cisco Router digunakan sebagai gateway untuk seluruh VLAN dengan metode *Router-on-a-Stick*. Satu interface fisik router dibuat menjadi beberapa subinterface, dan setiap subinterface diberi enkapsulasi VLAN serta IP gateway masing-masing.

1. Masuk ke Cisco Router melalui mode konfigurasi global.
2. Mengaktifkan interface fisik yang terhubung ke switch sebagai trunk tanpa IP address langsung.
3. Membuat subinterface VLAN 10, 20, 30, dan 40 dengan enkapsulasi dot1q.
4. Mengaktifkan DHCP Server untuk VLAN 10 dan VLAN 20.
5. Mengatur IP statis pada VPCS VLAN 30 dan VLAN 40.
6. Memverifikasi interface, DHCP pool, DHCP binding, serta pengujian ping gateway dan antar-VLAN.

```
interface gig0/2
  description TRUNK-TO-CISCO-SWITCH
  no ip address
  no shutdown
exit
```

Gambar 4: Dokumentasi konfigurasi Cisco Router 1

```
interface gig0/2.10
description GATEWAY-VLAN10-FINANCE
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
```

Gambar 5: Dokumentasi konfigurasi Cisco Router 2

```
interface gig0/2.20
description GATEWAY-VLAN20-HR
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
```

Gambar 6: Dokumentasi konfigurasi Cisco Router 3

```
interface gig0/2.30
description GATEWAY-VLAN30-IT
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
no shutdown
exit
```

Gambar 7: Dokumentasi konfigurasi Cisco Router 4

```
interface gig0/2.40
  description GATEWAY-VLAN40-MARKETING
  encapsulation dot1Q 40
  ip address 192.168.40.1 255.255.255.0
  no shutdown
  exit
```

Gambar 8: Dokumentasi konfigurasi Cisco Router 5

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10

ip dhcp pool VLAN10-FINANCE
  network 192.168.10.0 255.255.255.0
  default-router 192.168.10.1
  dns-server 8.8.8.8
  exit
```

Gambar 9: Dokumentasi konfigurasi Cisco Router 6

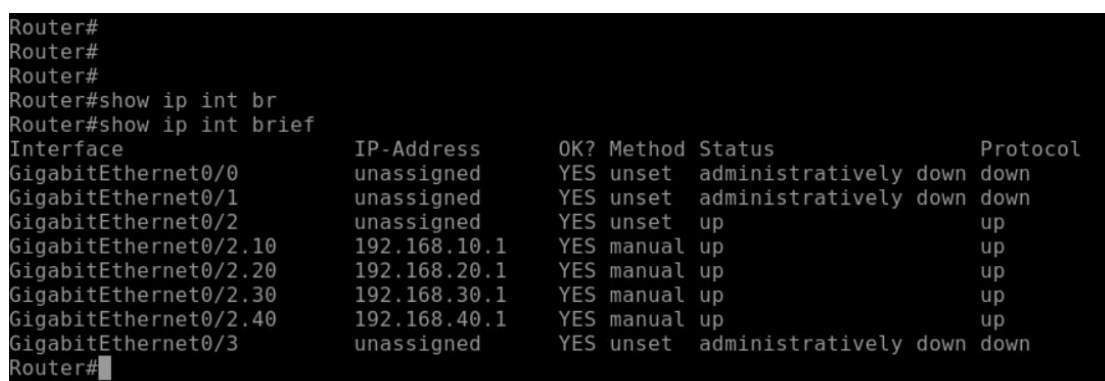
```
ip dhcp excluded-address 192.168.20.1 192.168.20.10

ip dhcp pool VLAN20-HR
  network 192.168.20.0 255.255.255.0
  default-router 192.168.20.1
  dns-server 8.8.8.8
  exit
```

Gambar 10: Dokumentasi konfigurasi Cisco Router 7



Gambar 11: Dokumentasi konfigurasi Cisco Router 8



Gambar 12: Dokumentasi konfigurasi Cisco Router 9

1.3 Konfigurasi Huawei Router dan IP Address Antarrouter

Percobaan ketiga menyiapkan alamat IP pada Huawei Router, Cisco Router, dan MikroTik sebelum OSPF dijalankan. Link antarrouter menggunakan subnet /30 karena setiap link hanya membutuhkan dua alamat host, sehingga alokasi IP lebih efisien.

1. Mengonfigurasi LAN Huawei pada network 192.168.50.0/24 dengan gateway 192.168.50.1.
2. Mengatur PC LAN Huawei dengan IP 192.168.50.10/24 dan gateway 192.168.50.1.
3. Mengonfigurasi dua link Cisco–Huawei: 10.10.10.0/30 dan 10.10.11.0/30.
4. Mengonfigurasi link Huawei–MikroTik pada 10.10.20.0/30 dan Cisco–MikroTik pada 10.10.30.0/30.
5. Melakukan verifikasi interface dan ping antarrouter.

```
system-view

interface Ethernet1/0/2
description LAN-HUAWEI
ip address 192.168.50.1 255.255.255.0
undo shutdown
quit

commit
```

Gambar 13: Dokumentasi konfigurasi Huawei dan antarrouter 1

Cisco Router

```
enable
configure terminal

interface gi0/3
description LINK1-TO-HUAWEI
ip address 10.10.10.1 255.255.255.252
no shutdown
exit
```

Huawei Router

```
system-view

interface Ethernet1/0/0
description LINK1-TO-CISCO
ip address 10.10.10.2 255.255.255.252
undo shutdown
quit

commit
```

Gambar 14: Dokumentasi konfigurasi Huawei dan antarrouter 2

Cisco Router

```
interface gi0/0
description LINK2-T0-HUAWEI
ip address 10.10.11.1 255.255.255.252
no shutdown
exit
```

Huawei Router

```
system-view

interface Ethernet1/0/4
description LINK2-T0-CISCO
ip address 10.10.11.2 255.255.255.252
undo shutdown
quit

commit
```

Gambar 15: Dokumentasi konfigurasi Huawei dan antarrouter 3

Huawei Router

```
system-view

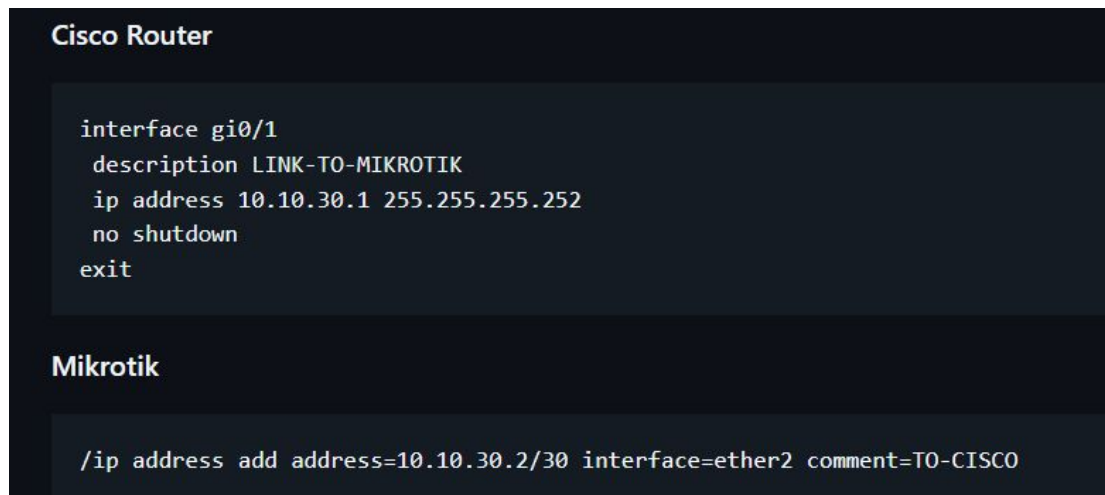
interface Ethernet1/0/3
description LINK-T0-MIKROTIK
ip address 10.10.20.1 255.255.255.252
undo shutdown
quit

commit
```

Mikrotik

```
/ip address add address=10.10.20.2/30 interface=ether1 comment=T0-HUAWEI
```

Gambar 16: Dokumentasi konfigurasi Huawei dan antarrouter 4



Gambar 17: Dokumentasi konfigurasi Huawei dan antarrouter 5

Verifikasi dilakukan dengan ping antar perangkat yang saling terhubung langsung. Cisco harus dapat menjangkau Huawei dan MikroTik, Huawei harus dapat menjangkau Cisco dan MikroTik, serta MikroTik harus dapat menjangkau Cisco dan Huawei.

1.4 Konfigurasi OSPF Multivendor dan OSPF Cost

Percobaan keempat menerapkan OSPF pada Cisco, Huawei, dan MikroTik. Cisco dan Huawei memiliki dua jalur utama dengan cost rendah, sedangkan MikroTik dijadikan jalur cadangan dengan cost lebih tinggi. Dengan pengaturan ini, OSPF akan memilih link Cisco–Huawei sebagai jalur utama dan menggunakan MikroTik ketika jalur utama gagal.

1. Pada Cisco Router, OSPF process 1 dikonfigurasi dengan router-id 1.1.1.1. Seluruh network VLAN dan link antarrouter diiklankan ke area 0.
2. Pada Huawei Router, OSPF process 1 dikonfigurasi dengan router-id 2.2.2.2. Network link dan LAN Huawei dimasukkan ke area 0.0.0.0.
3. Pada MikroTik, router-id diset menjadi 3.3.3.3, lalu network 10.10.20.0/30 dan 10.10.30.0/30 dimasukkan ke backbone.
4. Cost pada link utama Cisco–Huawei dibuat 10, sedangkan link via MikroTik dibuat 100 agar menjadi jalur backup.

```
router ospf 1
  router-id 1.1.1.1
  network 10.10.10.0 0.0.0.3 area 0
  network 10.10.11.0 0.0.0.3 area 0
  network 10.10.30.0 0.0.0.3 area 0
  network 192.168.10.0 0.0.0.255 area 0
  network 192.168.20.0 0.0.0.255 area 0
  network 192.168.30.0 0.0.0.255 area 0
  network 192.168.40.0 0.0.0.255 area 0
exit
```

Gambar 18: Konfigurasi OSPF pada Cisco Router

Konfigurasi OSPF:

```
ospf 1 router-id 2.2.2.2
 area 0.0.0.0
   network 10.10.10.0 0.0.0.3
   network 10.10.11.0 0.0.0.3
   network 10.10.20.0 0.0.0.3
   network 192.168.50.0 0.0.0.255
 quit
quit
```

Atur OSPF cost pada interface huawei.

```
interface Ethernet1/0/0
 description LINK1-TO-CISCO
 ospf cost 10
 quit

interface Ethernet1/0/4
 description LINK2-TO-CISCO
 ospf cost 10
 quit

interface Ethernet1/0/3
 description LINK-TO-MIKROTIK
 ospf cost 100
 quit

commit
```

Gambar 19: Konfigurasi OSPF pada Huawei Router

Pengujian end-to-end dilakukan dari setiap PC VLAN menuju PC LAN Huawei dan sebaliknya. Keberhasilan ping membuktikan bahwa route OSPF sudah tersebar ke semua router dan konektivitas antar network berjalan lintas vendor.

1.5 Pengujian Failover OSPF

Percobaan terakhir menguji kemampuan OSPF memilih jalur alternatif ketika link utama mengalami gangguan. Pengujian dilakukan dengan kondisi dua link Cisco–Huawei aktif, satu link dimatikan, dua link dimatikan, lalu link utama diaktifkan kembali.

1. Pada kondisi normal, traceroute menunjukkan trafik melewati jalur langsung Cisco–Huawei.
2. Ketika satu link Cisco–Huawei dimatikan, trafik tetap menggunakan link Cisco–Huawei lain yang masih aktif.
3. Ketika dua link utama dimatikan, jalur berpindah ke Cisco–MikroTik–Huawei sebagai jalur backup.
4. Setelah link utama diaktifkan kembali, OSPF memilih lagi jalur utama karena cost-nya lebih rendah.

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    2.664 ms  1.766 ms  1.699 ms
 2  10.10.10.2     5.802 ms  2.401 ms  2.825 ms
 3  *192.168.50.10 17.171 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS> █
```

Gambar 20: Pengujian failover OSPF: Kondisi normal

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    6.264 ms  2.149 ms  2.647 ms
 2  10.10.11.2     3.815 ms  2.709 ms  1.908 ms
 3  *192.168.50.10 14.584 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS> █
```

Gambar 21: Pengujian failover OSPF: Satu link mati

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    2.501 ms  2.305 ms  2.885 ms
 2  10.10.30.2     2.714 ms  1.933 ms  2.174 ms
 3  10.10.20.1     4.843 ms  2.507 ms  2.775 ms
 4  *192.168.50.10 3.970 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS> █
```

Gambar 22: Pengujian failover OSPF: Dua link mati melalui MikroTik

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    5.783 ms  8.876 ms  4.795 ms
 2  10.10.11.2     9.134 ms  2.152 ms  2.584 ms
 3  *192.168.50.10 10.945 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS> █
```

Gambar 23: Pengujian failover OSPF: Link utama aktif kembali

2 Analisis Hasil Percobaan

Konfigurasi VLAN pada Cisco Switch berhasil membagi jaringan fisik menjadi beberapa segmen logis. Perintah `show vlan brief` menunjukkan VLAN 10, 20, 30, dan 40 aktif, sedangkan konfigurasi access port memastikan setiap PC berada pada VLAN yang benar. Trunk menuju Cisco Router juga berhasil membawa beberapa VLAN melalui satu link menggunakan 802.1Q.

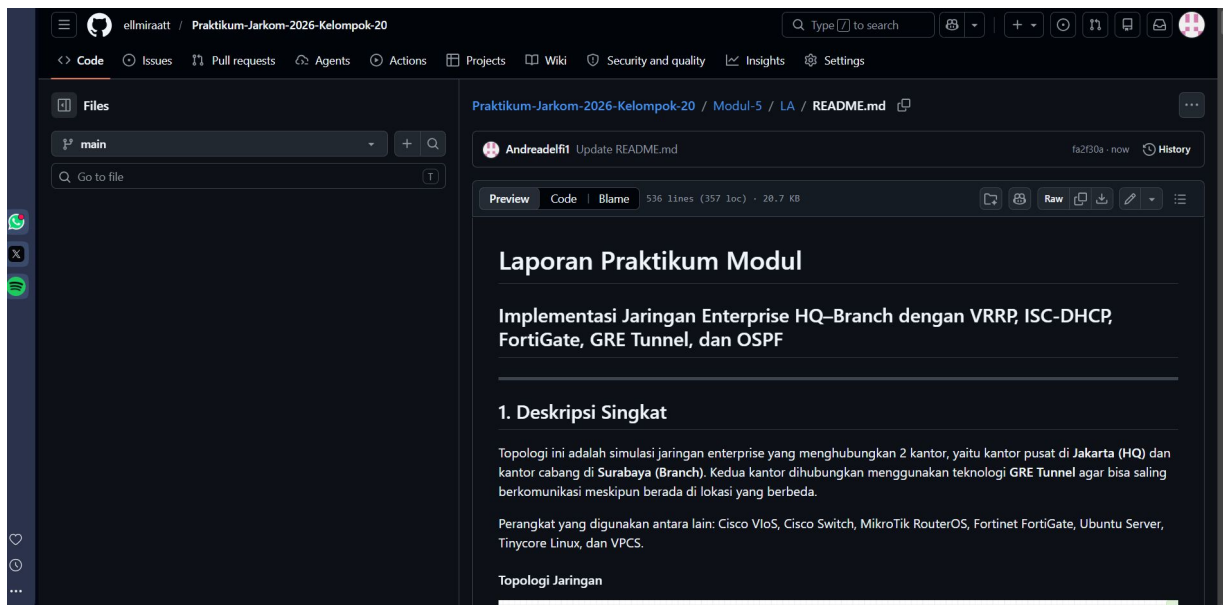
Implementasi Router-on-a-Stick pada Cisco Router berjalan sesuai teori. Setiap VLAN memiliki subinterface dan gateway masing-masing, sehingga proses routing antar-VLAN dapat dilakukan oleh satu interface fisik. DHCP Server untuk VLAN 10 dan VLAN 20 juga bekerja karena client memperoleh IP otomatis, sedangkan VLAN 30 dan VLAN 40 tetap dapat berkomunikasi menggunakan IP statis.

Konfigurasi IP antarrouter menggunakan subnet /30 terbukti tepat untuk link point-to-point. Seluruh link Cisco–Huawei, Huawei–MikroTik, dan Cisco–MikroTik dapat saling ping, sehingga fondasi routing dinamis sudah siap sebelum OSPF dijalankan.

Pada konfigurasi OSPF multivendor, Cisco, Huawei, dan MikroTik berhasil membentuk hubungan neighbor atau peer. Perbedaan vendor tidak menghalangi pertukaran routing karena OSPF merupakan protokol standar. Pengaturan cost juga memengaruhi pemilihan jalur, di mana link Cisco–Huawei menjadi jalur utama dan jalur melalui MikroTik menjadi cadangan.

Pengujian failover menunjukkan OSPF mampu melakukan reconvergence ketika link utama bermasalah. Saat satu link mati, trafik tetap menggunakan link utama yang tersisa. Saat dua link Cisco–Huawei mati, trafik berpindah ke jalur backup melalui MikroTik. Ketika link utama aktif kembali, OSPF kembali memilih jalur dengan cost paling rendah.

3 Hasil Tugas Modul



Gambar 24: Topologi tugas modul jaringan enterprise HQ–Branch

Tugas modul P5 menerapkan konsep jaringan enterprise HQ–Branch dengan VRRP, DHCP Server, FortiGate, GRE Tunnel, dan OSPF. Topologi ini menggambarkan kantor pusat Jakarta dan cabang Surabaya yang saling terhubung melalui jaringan ISP dan GRE Tunnel. Konsep yang digunakan

merupakan pengembangan dari praktikum utama karena menggabungkan segmentasi VLAN, *high availability* gateway, routing dinamis, serta koneksi antar site.

Konfigurasi lengkap, tabel *addressing*, serta hasil pengujian tugas modul telah didokumentasikan secara terperinci dalam file README.md pada *repository* GitHub kelompok di folder LA/.

4 Kesimpulan

Praktikum VLAN, trunk, OSPF multivendor, dan failover berhasil dilaksanakan sesuai tujuan. VLAN dan trunk membuktikan bahwa segmentasi jaringan dapat dilakukan secara logis tanpa memerlukan switch fisik terpisah. Router-on-a-Stick memungkinkan satu router menjadi gateway untuk banyak VLAN, sedangkan DHCP Server membantu pemberian IP otomatis pada client.

Konfigurasi antarrouter dengan subnet /30 berhasil membangun konektivitas dasar antardevice. OSPF pada Cisco, Huawei, dan MikroTik menunjukkan bahwa protokol routing standar dapat bekerja lintas vendor. Pengaturan cost berhasil mengarahkan trafik ke jalur utama dan menyediakan jalur cadangan melalui MikroTik. Pengujian failover membuktikan bahwa OSPF dapat menyesuaikan jalur secara otomatis ketika terjadi kegagalan link dan kembali memilih jalur terbaik saat link utama pulih.

5 Lampiran

5.1 Dokumentasi saat Praktikum

Berikut adalah dokumentasi tambahan dari proses konfigurasi dan pengujian selama praktikum.

```
Switch#show vlan brief
Switch#show vlan brief
VLAN Name                Status Ports
-----
1    default                active  Gi0/0, Gi0/1, Gi0/2, Gi0/3
10   FINANCE                 active  Gi1/0, Gi1/1, Gi1/2, Gi1/3
20   HR                      active
30   IT                      active
40   MARKETING               active
1002 fddi-default            act/unsup
1003 token-ring-default     act/unsup
1004 fddinet-default         act/unsup
1005 trnet-default          act/unsup
Switch#
```

```
Switch#show interfaces trunk
Port      Mode          Encapsulation  Status      Native vlan
Gi0/0     on            802.1q         trunking    1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,30,40
Switch#
```

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  unassigned      YES unset  administratively down  down
GigabitEthernet0/1  unassigned      YES unset  administratively down  down
GigabitEthernet0/2  unassigned      YES unset  up          up
GigabitEthernet0/2.10  192.168.10.1   YES manual up          up
GigabitEthernet0/2.20  192.168.20.1   YES manual up          up
GigabitEthernet0/2.30  192.168.30.1   YES manual up          up
GigabitEthernet0/2.40  192.168.40.1   YES manual up          up
GigabitEthernet0/3  unassigned      YES unset  administratively down  down
Router#
```

```
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
Hardware address/
User name
192.168.10.11    0100.5079.6668.84 Jun 06 2026 02:37 AM    Automatic
192.168.20.11    0100.5079.6668.83 Jun 06 2026 02:37 AM    Automatic
Router#
```

```
VPCS> show ip
NAME           : VPCS[1]
IP/MASK        : 192.168.10.11/24
GATEWAY        : 192.168.10.1
DNS            : 8.8.8.8
DHCP SERVER    : 192.168.10.1
DHCP LEASE     : 86231, 86400/43200/75600
MAC            : 08:50:79:66:68:84
I/PORNT        : 20000
RHOST:PORT     : 127.0.0.1:30000
MTU            : 1500
VPCS>
```

```
VPCS> ping 192.168.10.1
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.620 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=3.333 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=3.840 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=3.083 ms
VPCS>
```

```
VPCS> show ip
NAME           : VPCS[1]
IP/MASK        : 192.168.20.11/24
GATEWAY        : 192.168.20.1
DNS            : 8.8.8.8
DHCP SERVER    : 192.168.20.1
DHCP LEASE     : 86172, 86400/43200/75600
MAC            : 08:50:79:66:68:83
I/PORNT        : 20000
RHOST:PORT     : 127.0.0.1:30000
MTU            : 1500
VPCS>
```

```
VPCS> ping 192.168.20.1
84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=3.074 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=3.001 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=3.445 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=5.102 ms
VPCS>
```

```
VPCS> ping 192.168.30.1
84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=3.134 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=4.138 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=4.657 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=2.955 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=5.074 ms
^C
VPCS>
```

```
VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=5.493 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=6.770 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=5.525 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=6.121 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=5.673 ms
VPCS>
```

```
VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=5.860 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=5.465 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=4.598 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=6.202 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=5.523 ms
^[[A
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=5.181 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=5.263 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=4.893 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=5.073 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=5.731 ms
VPCS>
```

```
VPCS> ping 192.168.50.1
84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=9.769 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=3.222 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=3.317 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=3.757 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=3.085 ms
VPCS>
```

```
[Huawei@HUAWEI]display ip interface brief
down: administratively down
down: FIB overload down
down: standby
(l): loopback
s): spoofing
d): Dampening Suppressed
E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface      IP Address/Mask      Physical      Protocol VPN
Ethernet0/0    10.10.10.2/30         up            up      --
Ethernet0/0.4094 128.1.138.137/16     up            up      l3vpn
Ethernet0/1     unassigned            down         --
Ethernet0/1.4094 128.1.138.137/16     up            up      l3vpn
Ethernet0/2     192.168.50.1/24       up            up      --
Ethernet0/2.4094 128.1.138.137/16     up            up      l3vpn
Ethernet0/3     10.10.20.1/30         up            up      --
Ethernet0/3.4094 128.1.138.137/16     up            up      l3vpn
Ethernet0/4     10.10.11.2/30         up            up      --
Ethernet0/4.4094 128.1.138.137/16     up            up      l3vpn
---- More ----
```

```
VPCS> ping 192.168.40.1
84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=3.336 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=3.413 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=4.277 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=3.569 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=3.007 ms
VPCS>
```

```
VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1
VPCS> show ip
NAME      : VPCS[1]
IP/MASK   : 192.168.50.10/24
GATEWAY   : 192.168.50.1
DNS       :
MAC       : 00:50:79:66:68:85
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU       : 1500
VPCS>
```

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0 10.10.11.1     YES manual up          up
GigabitEthernet0/1 10.10.30.1     YES manual up          up
GigabitEthernet0/2 unassigned      YES unset  up          up
GigabitEthernet0/2.10 192.168.10.1  YES manual up          up
GigabitEthernet0/2.20 192.168.20.1  YES manual up          up
GigabitEthernet0/2.30 192.168.30.1  YES manual up          up
GigabitEthernet0/2.40 192.168.40.1  YES manual up          up
GigabitEthernet0/3 10.10.10.1     YES manual up          up
Router#
```

```
[admin@MikroTik] > ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ::: TO-HUAWEI 10.10.20.2/30 ether1
1 ::: TO-CISCO 10.10.30.2/30 ether2
[admin@MikroTik] >
```

```
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST                               SIZE TTL TIME STATUS
0 10.10.20.1                          56 255 6ms
1 10.10.20.1                          56 255 1ms
2 10.10.20.1                          56 255 1ms
3 10.10.20.1                          56 255 1ms
4 10.10.20.1                          56 255 1ms
5 10.10.20.1                          56 255 1ms
6 10.10.20.1                          56 255 1ms
7 10.10.20.1                          56 255 1ms
sent=8 received=8 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=6ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST                               SIZE TTL TIME STATUS
0 10.10.30.1                          56 255 5ms
1 10.10.30.1                          56 255 1ms
2 10.10.30.1                          56 255 1ms
3 10.10.30.1                          56 255 1ms
4 10.10.30.1                          56 255 1ms
5 10.10.30.1                          56 255 1ms
6 10.10.30.1                          56 255 1ms
sent=7 received=7 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=5ms

[admin@MikroTik] >
```

```
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
Router#
```

```
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

```
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

```
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 110)

Router#
```

```
(M) Indicates MADJ neighbor

OSPF Process 1 with Router ID 2.2.2.2
  Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
  Router ID: 1.1.1.1      Address: 10.10.10.1
  State: Full           Mode:Nbr is Slave   Priority: 1
  DR: 10.10.10.2        BDR: 10.10.10.1    MTU: 1500
  Dead timer due in 32 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m34s
  Neighbor Up Time : 2026-06-05 03:25:13
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
  Router ID: 1.1.1.1      Address: 10.10.11.1
  State: Full           Mode:Nbr is Slave   Priority: 1
  DR: 10.10.11.2        BDR: 10.10.11.1    MTU: 1500
  Dead timer due in 34 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m39s
  Neighbor Up Time : 2026-06-05 03:25:08
  Authentication Sequence: [ 0 ]

[~HUAWEI]
[~HUAWEI]
```

```
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public_Routing Table : OSPF
  Destinations : 9          Routes : 14

OSPF routing table status : <Active>
  Destinations : 5          Routes : 10

Destination/Mask Proto Pre Cost Flags NextHop Interface
-----
10.10.30.0/30 OSPF 10 110 D 10.10.11.1 Ethernet1/0/4
10.10.10.0/30 OSPF 10 110 D 10.10.10.1 Ethernet1/0/0
192.168.10.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
10.10.11.0/30 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.20.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
10.10.10.0/30 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.30.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
10.10.11.0/30 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.40.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
10.10.10.0/30 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0

OSPF routing table status : <Inactive>
  Destinations : 4          Routes : 4

Destination/Mask Proto Pre Cost Flags NextHop Interface
-----
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/0
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/4
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/3
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2

[~HUAWEI]
```

```
[~HUAWEI]display ospf routing

OSPF Process 1 with Router ID 2.2.2.2
  Routing Tables

Routing for Network
Destination Cost Type NextHop AdvRouter Area
-----
10.10.10.0/30 10 Direct 10.10.10.2 2.2.2.2 0.0.0.0
10.10.11.0/30 10 Direct 10.10.11.2 2.2.2.2 0.0.0.0
10.10.20.0/30 100 Direct 10.10.20.1 2.2.2.2 0.0.0.0
10.10.30.0/30 110 Stub 10.10.11.1 1.1.1.1 0.0.0.0
10.10.30.0/30 110 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.50.0/24 1 Direct 192.168.50.1 2.2.2.2 0.0.0.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0
---- More ----
```

```
[admin@Mikrotik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=0.0.0.0 state="ExStart" state-changes=3
ls-retransmits=0 ls-requests=0 db-summaries=0

1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=6s
[admin@Mikrotik] >
```

```
[admin@Mikrotik] > ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 Ado 10.10.10.0/30 10.10.30.1 110
1 Ado 10.10.11.0/30 10.10.30.1 110
2 Ado 192.168.10.0/24 10.10.30.1 110
3 Ado 192.168.20.0/24 10.10.30.1 110
4 Ado 192.168.30.0/24 10.10.30.1 110
5 Ado 192.168.40.0/24 10.10.30.1 110
6 Ado 192.168.50.0/24 10.10.20.1 110
[admin@Mikrotik] >
```

```
Router#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/BDR 00:00:37 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/DR 00:00:32 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/DR 00:00:37 10.10.10.2 GigabitEthernet0/3
Router#
```

```
~HUAWEI[display ospf peer
M) Indicates MADJ neighbor

OSPF Process 1 with Router ID 2.2.2.2
Neighbors
Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.2 BDR: 10.10.10.1 MTU: 1500
Dead timer due in 35 sec
Retrans timer interval: 5
Neighbor is up for 00h04m14s
Neighbor Up Time : 2026-06-05 03:25:13
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3 Address: 10.10.20.2
State: Full Mode:Nbr is Master Priority: 1
DR: 10.10.20.1 BDR: 10.10.20.2 MTU: 1500
Dead timer due in 32 sec
Retrans timer interval: 5
Neighbor is up for 00h01m05s
Neighbor Up Time : 2026-06-05 03:28:22
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
---- More ----
```

```
[admin@Mikrotik] > routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=1m54s

1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m3s
[admin@Mikrotik] >
```

```
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:05:09 ago
Routing Descriptor Blocks:
10.10.11.2, from 2.2.2.2, 00:05:09 ago, via GigabitEthernet0/0
Route metric is 11, traffic share count is 1
10.10.10.2, from 2.2.2.2, 00:05:09 ago, via GigabitEthernet0/3
Route metric is 11, traffic share count is 1
Router#
```

```
~HUAWEI[display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public Routing Table : OSPF
Destinations : 9 Routes : 14

OSPF routing table status : <Active>
Destinations : 5 Routes : 10

Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.30.0/30 OSPF 10 110 D 10.10.11.1 Ethernet1/0/4
10.10.30.0/30 OSPF 10 110 D 10.10.10.1 Ethernet1/0/0
192.168.10.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.10.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.20.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.20.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.30.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.30.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.40.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.40.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4 Routes : 4

Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/0
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/4
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/3
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2
~HUAWEI]
```

```
[admin@Mikrotik] > ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 Ado 10.10.10.0/30 10.10.30.1 110
1 Ado 10.10.11.0/30 10.10.30.1 110
2 Ado 192.168.10.0/24 10.10.30.1 110
3 Ado 192.168.20.0/24 10.10.30.1 110
4 Ado 192.168.30.0/24 10.10.30.1 110
5 Ado 192.168.40.0/24 10.10.30.1 110
6 Ado 192.168.50.0/24 10.10.20.1 110
[admin@Mikrotik] >
```

```
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=12.786 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=11.120 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.798 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.112 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=7.019 ms
VPCS>
```

```
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.197 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.257 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.816 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=9.935 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=8.487 ms
VPCS>
```

```
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=5.533 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=6.046 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=5.577 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=5.945 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=6.033 ms
VPCS>
```

```
VPCS>
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.234 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=5.390 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.306 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.030 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.855 ms
VPCS>
```

```
VPCS> ping 192.168.10.11
84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=7.906 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=5.975 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=8.083 ms
84 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=5.869 ms
84 bytes from 192.168.10.11 icmp_seq=5 ttl=62 time=9.098 ms
VPCS>
```

```
VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=7.081 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=10.277 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=8.245 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=62 time=6.950 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=62 time=8.759 ms
VPCS>
```

```
VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=6.784 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=6.938 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=6.912 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=9.874 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=7.791 ms
VPCS>
```

```
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=6.412 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=7.612 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=8.503 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=7.285 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=6.248 ms
VPCS>
```

```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.40.1 4.823 ms 13.168 ms 3.707 ms
 2 10.10.10.2 7.138 ms 4.418 ms 4.430 ms
 3 *192.168.50.10 11.531 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>
```

```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.40.1 4.012 ms 5.762 ms 4.780 ms
 2 10.10.11.2 9.748 ms 5.180 ms 3.660 ms
 3 *192.168.50.10 6.658 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>
```

```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.40.1 5.836 ms 3.170 ms 3.815 ms
 2 10.10.30.2 7.239 ms 3.662 ms 4.809 ms
 3 10.10.20.1 13.104 ms 0.254 ms 5.117 ms
 4 *192.168.50.10 8.631 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>
```

```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.40.1 5.164 ms 4.485 ms 3.399 ms
 2 10.10.10.2 6.618 ms 5.867 ms 4.695 ms
 3 *192.168.50.10 7.003 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>
```